- Suppose data from A fools B but data from B does not fool A. In this situation, we choose model A. It provides more information than model B. (If the converse is true, choose model B.)

- Suppose data from A does not fool B and data from B does not fool A. Then each model is capturing an aspect of the data that the other does not. Both models are valuable.

Enumerating these scenarios suggests a way to explore the differences between classes of models, particularly those that do not necessarily admit a natural ordering in terms of complexity.

The PPN study is related to a large literature on Bayesian predictive models for model criticism. A thorough review of such methods is provided by Vehtari and Ojanen (2012). In particular, a related method was developed in Gelfand and Ghosh (1998), which proposes to select a model that minimizes the expected error in predicting data from the posterior predictive distribution. The PPN builds on such predictive methods by determining whether the posterior predictive distribution of one model can fool the predictive check for another model. In this way, the PPN provides a notion of model similarity based on predictive distributions. Further, the PPN takes into account the sampling variability in the posterior predictive distribution.

2.4 Computing realized diagnostics

The diagnostic $d_{\mathrm{A}}(\mathbf{x})$ is a function that quantifies, in some way, how incompatible the data $\mathbf{x}$ are to model A. In designing diagnostics, it is often natural and convenient to consider function of the latent variables specified in the model. Gelman et al. (1996) refer to such functions as "realized" because they require a realization of the latent variables. For example, a common realized diagnostic is the negative log likelihood of the data,

$$d_{\mathrm{A}}(\mathbf{x}, \theta_{\mathrm{A}}) = -\log \mathrm{p}_{\mathrm{A}}(\mathbf{x} \,|\, \theta_{\mathrm{A}}), \tag{9}$$

where θ_{A} are the latent parameters of model A. Large values of this diagnostic mean the data are incompatible with the realization of the latent variable.

When we use a realized diagnostic, we have to decide how to handle the latent variable. One possibility is to remove it from the diagnostic, thereby forming a simple diagnostic from a realized one (Gelman et al., 1996). Examples of such diagnostics include the average and MAP diagnostic:

$$d_{\mathrm{A}}^{\mathrm{avg}}(\mathbf{x}) = \int d_{\mathrm{A}}(\mathbf{x}, \theta_{\mathrm{A}}) \mathrm{p}(\theta_{\mathrm{A}} \,|\, \mathbf{x}) \,\mathrm{d}\theta_{\mathrm{A}} \tag{10}$$

$$d_{\mathrm{A}}^{\mathrm{map}}(\mathbf{x}) = d_{\mathrm{A}}(\mathbf{x}, \theta_{\mathrm{A}}^{*}) \qquad \theta_{\mathrm{A}}^{*} = \arg\max_{\theta_{\mathrm{A}}} \log \mathrm{p}(\theta_{\mathrm{A}} \,|\, \mathbf{x}). \tag{11}$$

Note these diagnostics can be used in the context of a PPC or a PPN.

Such diagnostics, however, are still computationally expensive. To evaluate each one requires a minimization or posterior inference, and Bayesian model criticism tends to require many evaluations of the diagnostic, one for each replicate of the dataset.

To alleviate this burden, we propose a "validation diagnostic." The validation diagnostic marginalizes over the posterior of the latent parameters given a fixed held-out validation dataset $\mathbf{x}_{\mathrm{val}}$, one that is not used in the context of the model check. The validation diagnostic is

$$d_{\mathrm{A}}(\mathbf{x};\mathbf{x}_{\mathrm{val}}) = \int d_{\mathrm{A}}(\mathbf{x},\theta_{\mathrm{A}})\mathrm{p}_{\mathrm{A}}(\theta_{\mathrm{A}} \,|\, \mathbf{x}_{\mathrm{val}})\,\mathrm{d}\theta_{\mathrm{A}}. \tag{12}$$

This diagnostic avoids the computational cost of refitting the model to each replicated dataset.

In practice, we split the data into $\mathbf{x}_{\mathrm{obs}} = \{\mathbf{x}_{\mathrm{in}}, \mathbf{x}_{\mathrm{val}}\}$. We use the in-sample data to draw samples from the posterior predictive distribution. The diagnostic is then defined from the validation data. One might ask why not use the same data in both settings. The reason is that this would bias the diagnostic to favor the observed data, mirroring the "double counting" issue of the PPC. Specifically, the PPN with the validation diagnostic assesses the similarity of the distributions $\mathrm{p}_{\mathrm{A}}(d_{\mathrm{A}}(\mathbf{x}^{\mathrm{A}}_{\mathrm{rep}};\mathbf{x}_{\mathrm{val}}) \,|\, \mathbf{x}_{\mathrm{in}})$ and $\mathrm{p}_{\mathrm{B}}(d_{\mathrm{A}}(\mathbf{x}^{\mathrm{B}}_{\mathrm{rep}};\mathbf{x}_{\mathrm{val}}) \,|\, \mathbf{x}_{\mathrm{in}})$.

Note that when we use the PPN in concert with the heldout predictive check in Eq. 6, we instead split the data into three: $\mathbf{x}_{\mathrm{obs}} = \{\mathbf{x}_{\mathrm{in}}, \mathbf{x}_{\mathrm{out}}, \mathbf{x}_{\mathrm{val}}\}$, where

- $\mathbf{x}_{\mathrm{in}}$ is in-sample data used to draw from the posterior predictive distribution;
- $\mathbf{x}_{\mathrm{out}}$ is out-of-sample data located within the reference distribution for a heldout predictive check;
- $\mathbf{x}_{\mathrm{val}}$ is data used to calculate the validation diagnostic Eq. 12.

Then, the heldout predictive p-value with the validation diagnostic is:

$$\mathrm{P}_{\mathrm{hpred}}(d_{\mathrm{A}}(\mathbf{x}_{\mathrm{rep}};\mathbf{x}_{\mathrm{val}}) \geq d_{\mathrm{A}}(\mathbf{x}_{\mathrm{out}};\mathbf{x}_{\mathrm{val}})|\mathbf{x}_{\mathrm{in}}), \quad \mathbf{x}_{\mathrm{rep}} \sim \mathrm{p}_{A}(\mathbf{x}_{\mathrm{rep}}|\mathbf{x}_{\mathrm{in}}). \tag{13}$$

The heldout predictive check with the validation diagnostic is in Algorithm 1. A PPN with the validation diagnostic is in Algorithm 2. Finally, a PPN study, which combines predictive checks and PPNs, is in Algorithm 3.

2.5 A simple regression example

As a simple pedagogical example of a PPN, we compare two regression models for which the posterior predictive distributions are known exactly. The first "regression" $\mathcal{M}_A$ does not include any covariates, while the second $\mathcal{M}_B$ includes p covariates,

$$y_i|\theta, \mathcal{M}_A \sim N(\theta, 1), \quad p(\theta) \propto 1, \quad i = 1, \ldots, n \tag{14}$$

Algorithm 1: The heldout predictive check

input: data $\mathbf{x}_{\mathrm{obs}} = \{\mathbf{x}_{\mathrm{in}}, \mathbf{x}_{\mathrm{out}}, \mathbf{x}_{\mathrm{val}}\}$, model $\mathcal{M}_{\mathrm{A}}$ and diagnostic $d_{\mathrm{A}}(\cdot\ ; \mathbf{x}_{\mathrm{val}})$
output: heldout predictive check p-value
for $r = 1, \ldots, R$ **do**
  draw posterior predictive data $\mathbf{x}_{\mathrm{rep,r}} \sim \mathrm{P}(\mathbf{x}_{\mathrm{rep}} | \mathbf{x}_{\mathrm{in}}; \mathcal{M}_{\mathrm{A}})$
for $b = 1, \ldots, B$ **do**
  draw samples from the posterior $\theta_b \sim \mathrm{P}(\theta | \mathbf{x}_{\mathrm{val}}; \mathcal{M}_{\mathrm{A}})$
compute the empirical heldout predictive check p-value:

$$p_{hpred} = \frac{1}{R} \sum_{r=1}^{R} 1\left[d_{\mathrm{A}}(\mathbf{x}_{\mathrm{rep,r}}; \mathbf{x}_{\mathrm{val}}) > d_{\mathrm{A}}(\mathbf{x}_{\mathrm{out}}; \mathbf{x}_{\mathrm{val}})\right]$$

$$\text{where} \quad d_{\mathrm{A}}(\mathbf{x}; \mathbf{x}_{\mathrm{val}}) = \frac{1}{B} \sum_{b=1}^{B} d_{\mathrm{A}}(\mathbf{x}, \theta_b);$$

return p_{hpred}

Algorithm 2: The posterior predictive null

input: data $\mathbf{x}_{\mathrm{obs}} = \{\mathbf{x}_{\mathrm{in}}, \mathbf{x}_{\mathrm{out}}, \mathbf{x}_{\mathrm{val}}\}$, models $\mathcal{M}_{\mathrm{A}}$ and $\mathcal{M}_{\mathrm{B}}$ which pass their PCs, and diagnostic $d_{\mathrm{A}}(\cdot\ ; \mathbf{x}_{\mathrm{val}})$
output: $\mathrm{PPN}(d_{\mathrm{A}}, \mathrm{p}_{\mathrm{A}}, \mathrm{p}_{\mathrm{B}}, \mathbf{x}_{\mathrm{obs}})$
for $r = 1, \ldots, R$ **do**
  draw posterior predictive data $\mathbf{x}^{\mathrm{A}}_{\mathrm{rep,r}} \sim \mathrm{P}_{\mathrm{A}}(\mathbf{x}_{\mathrm{rep}} | \mathbf{x}_{\mathrm{in}}; \mathcal{M}_{\mathrm{A}})$
for $r = 1, \ldots, R$ **do**
  draw posterior predictive data $\mathbf{x}^{\mathrm{B}}_{\mathrm{rep,r}} \sim \mathrm{P}_{\mathrm{A}}(\mathbf{x}_{\mathrm{rep}} | \mathbf{x}_{\mathrm{in}}; \mathcal{M}_{\mathrm{B}})$
for $b = 1, \ldots, B$ **do**
  draw samples from the posterior $\theta_b \sim \mathrm{P}(\theta | \mathbf{x}_{\mathrm{val}}; \mathcal{M}_{\mathrm{A}})$
compute the empirical PPN

$$\mathrm{PPN}(d_{\mathrm{A}}, \mathrm{p}_{\mathrm{A}}, \mathrm{p}_{\mathrm{B}}, \mathbf{x}_{\mathrm{obs}}) = D\left(\left\{d_{\mathrm{A}}(\mathbf{x}^{\mathrm{A}}_{\mathrm{rep,r}}; \mathbf{x}_{\mathrm{val}})\right\}_{r=1}^{R} \,\middle\|\, \left\{d_{\mathrm{A}}(\mathbf{x}^{\mathrm{B}}_{\mathrm{rep,r}}; \mathbf{x}_{\mathrm{val}})\right\}_{r=1}^{R}\right)$$

$$\text{where} \quad d_{\mathrm{A}}(\mathbf{x}; \mathbf{x}_{\mathrm{val}}) = \frac{1}{B} \sum_{b=1}^{B} d_{\mathrm{A}}(\mathbf{x}, \theta_b);$$

return $PPN(d_{\mathrm{A}}, \mathrm{p}_{\mathrm{A}}, \mathrm{p}_{\mathrm{B}}, \mathbf{x}_{\mathrm{obs}})$